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Key To The Extraterrestrial Messages

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Dr. Campaigne presented a series of 29 messages from outer space in "Extraterrestrial Intelligence," NSA Technical Journal, Vol. XI, No. 2, pp. 101 ff. and in the Special Mathematics and Engineering Issue of the Journal, pp. 117 ff. The following article develops a key to these messages. Paragraph numbers parallel the serial numbers of the messages reprinted in the appendix below. This includes two new series--30 and 31--not included in the previous article.

At every step in the solution we make a guess at the meaning. Evidence will quickly accumulate to verify or refute this guess. The possibility of ambiguity of two consistent solutions is very remote. Only in the last steps, where verification is thin, could this happen.

1. There are 21 symbols, in the order given by this message.
 2. B is equivalent to AA, C to AAA, etc. That is, A=1; B=2; C=3; D=4; E=5; F=6; G=7.
 3. The symbol L means the two things that follow are the same. LXY means $x=y$.
 4. Each statement has 5 symbols, and begins with L. The 4 symbols after L must be considered as two things. Each statement has a K as the third letter, which must be the start of the second thing. Is $B=KAA$; $C=KBA$; $C=KAB$; $D=KCA$? If KBA means $B+A$, it fits.
 5. These verify our conclusions on 4. The first means $6=1+(2+3)$, the last means $1+(2+3)=(1+2)+3$.
 6. Each has five symbols as in 4. They mean $1=M21$; $2=M31$; $1=M32$. Obviously MXY means $x-y$.
 7. These translate $N=1-1$; $N=2-2$; $N=3-3$. N stands for zero, 0.
 8. These translate $1=O11$; $0=O01$; $2=O12$; $2=O21$; $0=O02$; $0=O20$; $4=O22$, etc. OXY means the product $X \times Y$.
 9. These verify the conclusions of 8. The first says that $6=1 \times 2 \times 3$, the last $4 \times (5 \times 6) = (4 \times 5) \times 6$.
- Note:* So far we have seen two kinds of symbols: digits A through G and N, and operators L, K, M, and O. The two digits following the operator are the operands.
10. Translates into $4=R22$; $2=R21$; $1=R20$; $3=R31$; $1=R30$. RXY must mean X^Y , exponentiation. R is another binary operator.

11. Translates into $2^1 = 4 + 4$; $2^2 = 1 + 7$; $2^3 = 2 \times 4$; $2^4 = 4$, verifying our previous conclusions.

12. Translates into $3^2 = 4 + 5$; $3^3 = 3 \times 3$. Further verification.

Note: In our culture we use parentheses to group closely associated terms, and as a first step it helps, even though it is not necessary, to put in parentheses. To do so unambiguously, start at the right and read left to the first operator symbol; put parentheses about the operator and the two quantities to its right. Repeat until no pair of parentheses contains more than an operator and two quantities.

13. Translates into $J = 2^4$; $J = 1 + 7$; $J + 1 = 3 \times 3$; $J + 1 = 3^2$, therefore $J = 8$.

14. We can only introduce parentheses by assuming P is an operator, so we get 2 P84; 4 P82; 1 P53; and 3 P62. Thus $P \times Y = X \div Y$, division.

15. Assume U is an operator, getting U12; U23; U34; U58; U68; U78; U234 = U89. The smaller is first in each case; so perhaps UXY means X precedes Y or $X < Y$.

16. The new character Q must be an operator. Transcribed it gives us Q: 0=1; Q: 1=3; Q: 1=1+1; Q: 0=1×1; Q: 2²=2; Q: 2<2; Q: H(1-1)(1-2); Q|Q: 3=3; Q: 8<7.

Clearly Q means "the following statement is false." Then the next to the last is read "it is false that $3=3$." Q1. will be translated $\neq$. The second new symbol is not clear, except that it is an operator whose operands are statements, not quantities, a Boolean operator.

Note: Q is an operator with only one operand, unary.

17. Putting in parentheses shows that S is also a unary operator operating on statements. Transcribed they are: S: 1=1; S: 1=2; S: 2=1+1; S: 1=1×1; S: 0=1-1; S: H[(6-1×6)(1-6:6)]; S: H[(1-1)(2-2)]. It is apparent that S means "the following statement is true" or "it is asserted that." The next to the last message shows that HXY means "X implies Y" or "X is a consequence of Y" or maybe "X is logically equivalent to Y."

18. Our rule for parentheses breaks down unless T is a different kind of symbol. The first message shows that T may be a unary operator on quantities, so that AT or TA is a quantity. The third message shows that it must be the first, since T is last. Putting in parentheses this way gives 1T=1; 2T=1T+1; 5T=2T+3T; 6T=3T×2T; 7T=7; 10T=8. T must be an ending. On one digit it makes no difference. It combines the two digits 10 to make 8. Octal arithmetic?

19. Transcribes to 123T = $1 \times 8^2 + (2 \times 8^1 + 3)$;

321T = $3 \times 8^2 + (2 \times 8^1 + 1 \times 8^0)$;

4567T = $(4 \times 8^3 + 5 \times 8^2) + (6 \times 8^1 + 7 \times 8^0)$.

Clearly T indicates that "the preceding digits form an octal number." Possibly it is an octal point; if so, digits may occur after it.

Note: Because of the way grouping is implied, it is sufficient to have a marker at the end of a number in order to clearly isolate it as a single entity.

20. In trying to put on parentheses it appears that V is also an ending. But this one combines with both quantities (that is, digits.) and operator. Transcribing by treating V and the preceding symbol as a single unit for the time being, we get:

$$8 \quad 1 + AV \text{ implies } 7 = AV.$$

I will use $\cdot$ for Π hereafter. Remember that we are not sure of the sense of this sign. $11 \cdot 3^{11} = \cdot 2 = AV$ (I have omitted the T. Remember that $11T$ is nine):

$$3 \quad AV = \cdot 11 = \overline{AV}^2;$$

$$5 \quad AV = \cdot 6 = 1 + AV.$$

In the next message if we combine the O and V into one symbol the message does not parse. Try GOV as one symbol, getting

$$3 \quad 1 + GOV = \cdot 2 = GOV;$$

$$6 \quad 2 \times SOV = \cdot 3 = SOV;$$

$$3 \quad DOV = \cdot 11 = \overline{DOV}^2;$$

$$\text{It is true that } AV \mid PV \mid = PV + AV;$$

$$\text{It is a tautology that } AV \times PV = PV \times AV;$$

$$\text{It is an identity that } AV + (PV + TV) = (AV + PV) + TV;$$

$$\text{It is asserted that } AV \times (PV \times TV) = (AV + PV) + TV;$$

$$AV \neq BV = \cdot AV = BV \neq BV \quad AV;$$

$$AV \neq BV = AV^{11} \neq BV^{11}.$$

The meaning of V must be that "the preceding letters as a group have an abstract meaning, or are a variable." V is a little like a word spacer.

Note: Putting in parentheses is now complicated by another rule. Each T or V should be packaged with preceding symbols, just how many depending on the parsing of the message. Those preceding T will all be digits. Those preceding V can be expected to reoccur as a group.

21. Putting parentheses in these messages is difficult until we notice that UV appears in each. They then transcribe into:

$$0 \quad [2 + (\overline{DDVV}^2 - DDVV \times 3)] = \cdot UV[1 = DDVV]$$

$$[2 = DDVV];$$

$$UV[1 = DV][0 = 1 = DV] = \cdot [1 = \overline{DV}^2];$$

$$UV[1 = BV][0 = BV] = \cdot 0 = BV - BV^2;$$

$$\text{It is true that } UV[AV = BV][AV \neq BV];$$

$$\text{It is true that } UV[AV < BV][BV < AV].$$

In order to complete the parsing we had to assume that UV was a binary operator, and in every case the operands are statements. It is clear from the algebra that UV means "or." The last message shows that U means $\neq$, rather than $<$ as I had it.

22. We notice that TV is used in every message, and parallels the usage of UV. Assuming TV is a binary Boolean operator, the messages parse.

It is false that $TV[AV \leq BV][BV \leq AV]$;

It is false that $TV[AV = BV][AV \neq BV]$;

$TV[AV = 4][0 \leq AV] = \cdot AV = 2$;

$AV > BV \Rightarrow \cdot \{BV \leq AV\} \text{ or } \{BV = AV\}$;

It is true that not $TV \text{ GV HV} = \cdot \text{GV or HV}$;

It is true that $GV \text{ and } HV = \cdot TV \text{ GV HV}$;

$AV \text{ or } (BV \text{ or } CV) = (AV \text{ or } BV) \text{ or } CV$;

$TV[AV \text{ TV}[BVCV]] = TV[TV \text{ AV BV}][CV]$;

$TV[AV[BV \text{ or } CV]] = TV[AVBV] \text{ or } TV[AV \text{ CV}]$;

$AV \text{ or } TV[BV \text{ CV}] = TV[AV \text{ or } BV][AV \text{ or } CV]$.

It is apparent that TV means "and." Notice that L is used here to mean "logically equivalent to," although I have written " $=$ ".

Note: U is used here for $<$, not $\leq$.

Either there is a mistake, or the usage varies.

23. The parsing falters until we realize that JNV occurs in each message, and is probably a word. BAV and CAV also occur in each message. They transcribe into:

$JNV[BAV \text{ or } CAV]BAV$;

$JNVBAV[BAV \text{ and } CAV]$;

$JNV[BAV \text{ or } CAV][BAV \text{ and } CAV]$;

$JNVBAVCAV = \cdot BAV = (BAV \text{ or } CAV)$;

$JNVBAVCAV = \cdot CAV = (BAV \text{ and } CAV)$.

The last two conclusions look like set theory statements. JNV parses like a binary operator. $JNVXY$ could mean "X contains Y" in the set theory sense. Then if UV is "or" in the set theory sense, the union, and TV is "and" in the set theory sense, the intersection, the statements above can be rewritten:

$BAV \cup CAV \supset BAV$

$BAV \supset BAV \cap CAV$

$BAV \cup CAV \supset BAV \cap CAV$

$BAV \supset CAV = \cdot BAV = (BAV \cup CAV)$

$BAV \supset CAV = \cdot CAV = (BAV \cap CAV)$.

24. NKV looks like a binary operator of which at least the first operand is a quantity. JAV is uniformly the second operand. From 23 above we are alert to set theory statements. Could it be that NKV says something is a member of some set? Try it. They become

$1cJAV; 2cJAV; 3cJAV; 4cJAV; 5cJAV; 6cJAV; 7cJAV; 11cJAV;$

$12cJAV; AVcJAV = \cdot AV + 1cJAV.$

JAV is the set of positive integers! It fits!

25. These parse into:

(1 and 2)₁ JAV;

(11 and 21 and 31)₁ JAV;

(14 and 17)₁ JAV;

(177 and 100) and (101)₁ JAV;

$(AV \supset NMV)$ and $(BV \supset NMV) = \cdot (AV \text{ and } BV) \supset NMV$;

0_1 JAV;

8_1 JAV; 8^2_1 JAV; 8^3_1 JAV; 8^4_1 JAV; 8^5_1 JAV; 8^6_1 JAV; 8^7_1 JAV;

8^{100}_1 JAV; BV_1 JAV $= \cdot 8^{100}_1$ JAV;

$(BV \text{ and } CV)_1$ JAV $= \cdot BV_1 CV_1$ JAV;

$(BV \text{ and } CV)_1$ JAV $= \cdot BV \times CV_1$ JAV;

$(BV \text{ and } CV)_1$ JAV $= \cdot BV^{100}_1$ JAV;

$1/2_1$ JAV; $1 \cdot 2_1$ JAV; $0 - 3_1$ JAV; $7/6_1$ JAV.

This verifies beyond doubt the guess of 24.

26. There is a new word, JOV. The messages read JAV₁JOV:

$0 \cdot 1_1$ JOV; $0 \cdot BV_1$ JAV $= \cdot BV_1$ JOV;

$1/2_1$ JOV; AV and BV_1 JOV $= \cdot AV - BV_1$ JOV;

(AV and BV in JOV) and $0 \neq BV = \cdot AV \div BV$ in JOV;

$1 \neq 0$ not in JAV; $1 \neq 0$ not in JOV;

It is true that $(AV \div BV) \times (CV \div DV) = (AV \times CV) \div (BV \times DV)$;

It is true that $AV \times DV < BV \times CV = \cdot AV \div BV < CV \div DV$, $BV \times DV \neq 0$;

AV_1 JAV $\cdot 0 \cdot 1 < AV$.

JOV is seen to be the field generated by JAV, in other words, the set of rational numbers. The next to the last message has a garble, an extraneous A.

27. This transcribes to:

$(AV = BV)$ and $(BV = \cdot AV) = \cdot HV.AV.BV$.

Clearly HV means "logically equivalent," or " $\cdot \cdot \cdot$ ".

$(AV = \cdot BV) = \cdot (AV = \cdot BV)$ and $(BV = \cdot AV)$

$(AV = \cdot BV) = \cdot (AV = \cdot BV)$ and $(BV = \cdot AV)$.

28. These transcribe to

$GV^1 = 3 \cdot \cdot \cdot$ GV not in JOV;

$GV^2 = 2 \cdot \cdot \cdot$ GV not in JOV;

$GV^3 = 5 \cdot \cdot \cdot$ GV not in JOV;

$GV^4 = 5 \cdot \cdot \cdot$ GV in JEV;

JOV is in JEV;

JAV is in JEV;

$GV^5 = 0 - 1 = \cdot$ GV not in JEV.

We have a new set, containing the rationals, and at least one irrational, but not the imaginary $\sqrt{-1}$. JEV is probably the real numbers.

29. These transcribe to

$1 \cdot 2^{\infty}$ in JBV; $1 \cdot 3^{\infty}$ in JBV;

$1 \cdot 4^{\infty}$ in JBV; $1 \cdot \bar{N}V^{\infty}$ in JBV;

NLV JBV = 1 [assuming NLV is one word. Another possible parsing is $1.V(JBV, 1) = 0$]

$1/1$ in JCV; $1/2$ in JCV; $1/3$ in JCV; $1/NV$ in JCV; NLV JCV = 0 [or $1.V(JCV, 0) = 0$. But the two examples suggest that NLV means "a limit of." If NV is an integer this fits perfectly.]

$(1 - 1/2)^{\infty}$ in JBV; $(1 - 1/3)^{\infty}$ in JBV;

$(1 - 1/4)^{\infty}$ in JBV; $(1 - \frac{1}{8^{100}})^{\infty}$ in JBV;

$(1 - \frac{1}{NV})^{\infty}$ in JBV; NLV JBV in JEV.

If NLV means limit, then JEV contains the number e, a verification of our guess that JEV named the real numbers.

The last two lessons - 30 and 31 - were not published with the first twenty-nine because it made too long an exercise.

30. The later messages of this group have the mysterious sequences ABCD, ABCDE, DEFG, etc., each ending with STV. If we hunch these each as a unit, the messages parse. They then say JNV 1 natural number; JNV 2 natural number; JNV 3 natural number; JNV 123 STV natural number; conjecture STV means "the preceding is a set (or sequence)," and JNV means "belongs to." There is doubt about the latter, since we thought earlier that it meant "contains"; AV belongs to 1234 = AV is a natural number; 12345 or 4567 = 45 as sets; 12345 and 4567 = 1234567 as sets.

31. This last group is of impressive magnitude, 41 messages, of which the thirtieth is quite long. Parsing is eased by the parallel construction of the messages. They transcribe to:

JRAV belongs to CHAV; JRBV belongs to CHAV; JRCV belongs to CHAV; the set JRAV, JRBV, JRCV, JRDV, JREV, JRFV, JRGV belongs to CHAV; [Since all the digits appear in these groups, maybe they are used like subscripts and should be read JR₁, JR₂, etc.]; JO₁ belongs to CHAV; JO₂ belongs to CHAV; JO₃ belongs to CHAV; the set JO₁, JO₂, JO₃, JO₄, JO₅, JO₆, JO₇, JO₈, JO₉, JO₁₀, JO₁₁, JO₁₂, JO₁₃, JO₁₄, JO₁₅, JO₁₆, JO₁₇, JO₁₈, JO₁₉, JO₂₀ belongs to CHAV; U₁ and U₂ = 22 JO₁ belongs to CHAV [This one must be parsed wrong or garbled]; BL₁ belongs to JR; BL₂ belongs to JR; BL₃ belongs to JR; BL₄ belongs to JR; BL₅ belongs to JR;

AV < 3 and 12 ≥ AV = BL_{1V} belongs to JR;

AV < 13 and 22 < AV = BL_{1V} belongs to JR;

AV < 23 and 44 < AV = BL_{1V} belongs to JR;

AV = 45 and 66 = AV = ... BL₁₁ belongs to JR₁;
 AV = 67 and 126 = AV = ... BL₁₁ belongs to JR₂;
 AV = 127 and 142 = AV = ... BL₁₁ belongs to JR₃;
 The set BL₁₁, BL₁₂, BL₁₃, BL₁₄, BL₁₅, BL₁₆, BL₁₇ belongs to JO₁;
 The set BL₁₈, BL₁₉, BL₂₀, BL₂₁, BL₂₂, BL₂₃, BL₂₄ belongs to JO₂;
 The set BL₂₅, BL₂₆, BL₂₇, BL₂₈, BL₂₉ belongs to JO₃;
 The set BL₃₀ and BL₃₁ and BL₃₂ and BL₃₃ and BL₃₄ belongs to JO₄;
 The set BL₃₅, BL₃₆, BL₃₇, BL₃₈ belongs to JO₅;
 The set BL₃₉, BL₄₀, BL₄₁, BL₄₂, BL₄₃ belongs to JO₆ [note a garble here, an N is repeated];
 The set BL₄₄, BL₄₅, BL₄₆, BL₄₇, BL₄₈ belongs to JO₇;
 The set BL₄₉, BL₅₀, BL₅₁, BL₅₂ belongs to JO₈;
 The set BL₅₃, BL₅₄, BL₅₅, BL₅₆, BL₅₇, BL₅₈, BL₅₉, BL₆₀, BL₆₁, BL₆₂, BL₆₃, BL₆₄, BL₆₅, BL₆₆, BL₆₇, BL₆₈, BL₆₉, BL₇₀, BL₇₁, BL₇₂, BL₇₃, BL₇₄, BL₇₅, BL₇₆, BL₇₇, BL₇₈, BL₇₉, BL₈₀, BL₈₁, BL₈₂, BL₈₃, BL₈₄, BL₈₅, BL₈₆, BL₈₇, BL₈₈, BL₈₉, BL₉₀, BL₉₁, BL₉₂, BL₉₃, BL₉₄, BL₉₅, BL₉₆, BL₉₇, BL₉₈ belongs to JO₉;
 BL₉₉ and BL₁₀₀ and BL₁₀₁ belongs to JO₁₀;
 The set BL₁₀₂, BL₁₀₃, BL₁₀₄ belongs to JO₁₁;
 The set BL₁₀₅, BL₁₀₆, BL₁₀₇ belongs to JO₁₂;
 The set BL₁₀₈, BL₁₀₉, BL₁₁₀ belongs to JO₁₃;
 The set BL₁₁₁, BL₁₁₂, BL₁₁₃ belongs to JO₁₄;
 The set BL₁₁₄, BL₁₁₅, BL₁₁₆ belongs to JO₁₅;
 The set BL₁₁₇, BL₁₁₈, BL₁₁₉ belongs to JO₁₆;
 The set BL₁₂₀, BL₁₂₁, BL₁₂₂ belongs to JO₁₇;
 The set BL₁₂₃, BL₁₂₄, BL₁₂₅ belongs to JO₁₈;
 The set BL₁₂₆, BL₁₂₇, BL₁₂₈ belongs to JO₁₉;
 The set BL₁₂₉, BL₁₃₀, BL₁₃₁ belongs to JO₂₀;
 CHAV belongs to KSPV.

The transcription leaves a lot to be resolved. There are several words the meanings of which are yet to be determined. The word CHAV (or CH) seems to be central. There are seven words JR, and eighteen words JO, and each of these belongs to CHAV. There are 98 words BL, each of which seems to belong to a unique JO. Does each also belong to a unique JR? With this hint we can straighten out the garbled message above: it reads "0 < AV and AV < 22 = . JO₁₁ belongs to CHAV"; there was a V omitted. I was also able to reparse six other messages. I will not bore you with the details, since the list above has been corrected.

Since each BL_i belongs to one JR, and JO_i, these can be displayed in a matrix

	JR ₁	JR ₂	JR ₃	JR ₄	JR ₅	JR ₆	JR ₇
JO ₁	BL ₁₁	BL ₁₂	BL ₁₃	BL ₁₄	BL ₁₅	BL ₁₆	BL ₁₇
JO ₂	BL ₁₈	BL ₁₉	BL ₂₀	BL ₂₁	BL ₂₂	BL ₂₃	BL ₂₄
JO ₃		BL ₂₅	BL ₂₆	BL ₂₇	BL ₂₈	BL ₂₉	
JO ₄		BL ₃₀	BL ₃₁	BL ₃₂	BL ₃₃	BL ₃₄	
JO ₅		BL ₃₅	BL ₃₆	BL ₃₇	BL ₃₈	BL ₃₉	
JO ₆		BL ₄₀	BL ₄₁	BL ₄₂	BL ₄₃	BL ₄₄	

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JO ₇	BL ₄₁	BL ₄₂	BL ₄₃	BL ₄₄	BL ₄₅	BL ₄₆	BL ₄₇	BL ₄₈	BL ₄₉	BL ₅₀
JO ₁₀	BL ₅₁	BL ₅₂	BL ₅₃	BL ₅₄	BL ₅₅	BL ₅₆	BL ₅₇	BL ₅₈	BL ₅₉	BL ₆₀
JO ₁₁	BL ₆₁	25	47	71	72	73	74	75	76	77
	100	101	102	103	104	105	106	107		
		131	132	133	134	135	136	137		
	140	141	142							
JO ₁₂	BL ₆₂				BL ₆₃	BL ₆₄	BL ₆₅	BL ₆₆	BL ₆₇	BL ₆₈
JO ₁₃				BL ₆₉	BL ₇₀	BL ₇₁	BL ₇₂	BL ₇₃	BL ₇₄	BL ₇₅
JO ₁₄				BL ₇₆	BL ₇₇	BL ₇₈	BL ₇₉	BL ₈₀	BL ₈₁	BL ₈₂
JO ₁₅				BL ₈₃	BL ₈₄	BL ₈₅	BL ₈₆	BL ₈₇	BL ₈₈	BL ₈₉
JO ₁₆				BL ₉₀	BL ₉₁	BL ₉₂	BL ₉₃	BL ₉₄	BL ₉₅	BL ₉₆
JO ₁₇				BL ₉₇	BL ₉₈	BL ₉₉	BL ₁₀₀	BL ₁₀₁	BL ₁₀₂	BL ₁₀₃
JO ₂₀				BL ₁₀₄	BL ₁₀₅	BL ₁₀₆	BL ₁₀₇	BL ₁₀₈	BL ₁₀₉	BL ₁₁₀
JO ₂₁				BL ₁₁₁	BL ₁₁₂	BL ₁₁₃	BL ₁₁₄	BL ₁₁₅	BL ₁₁₆	BL ₁₁₇
JO ₂₂				BL ₁₁₈	BL ₁₁₉	BL ₁₂₀	BL ₁₂₁	BL ₁₂₂	BL ₁₂₃	BL ₁₂₄

Remember that these are not decimal numbers. There is only one cell with more than one entry, and the subscripts in it in decimal notation are 21, 39, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98. The larger part of the entries is systematically distributed in the first eight rows. This suggests the periodic table of the chemical elements! On consulting a table we find, sure enough, that elements 57 through 71 are rare earths, and are lumped into one cell. Some, but not all, authorities also list 89 through 103 as rare earths. Elements 21 and 39 are Scandium and Yttrium.

CHAV must mean the periodic table. JR_x means column x, and JO_y means row y. BL_z means element Z. The meaning of KSPV is not known, except that it is a generalization of "periodic table." It may merely mean table, or scientific fact, or university subject.

Looking back over the exercise we see we have penetrated the meaning of the basic symbols, and even more important, have learned some of the syntax rules of the notation, and have caught mistakes in the process. We have a few words for sophisticated concepts, and, given more data, with a little labor we could establish its translation.

The concepts used here are the basic ones of number, sets, and physical constants which any cultures must share. How bizarre the syntax and values of a culture could be I cannot conjecture, but any civilizations capable of sending a message across space must have many things in common.

Recently a series of radio messages was heard coming from outer space. The transmission was not continuous, but cut by pauses into pieces which could be taken as units, for they were repeated over and over again. The pauses show here as punctuation. The various combinations have been represented by letters of the alphabet, so that the messages can be written down. Each message except the first is given here only once. The serial number of the message has been supplied for each reference.

- 1 ABCDEF GHIJKLMNOPQRSTU
ABCDEF GHIJKLMNOPQRSTU
ABCDEF GHIJKLMNOPQR etc
- 2 AA. B AAA.C. AAAA.D. AAAAA.E. AAAAAA.F. AAAAAAA.G
- 3 LAA. LBB. LCC. LDD. LEE. LFF. LGG
- 4 LBKAA. LBKBA. LBKAB. LBKCA. LBKCB. LBKAC. LBKDA.
LBKDB. LBKBC. LBKAD. LBKEA. LBKDB. LBKCC. LBKBD.
LBKAE
- 5 LFKAKHE. LFKCKHA. LFKAKBD. LOKCKAC. LKAKBCKKABC
- 6 LANHA. LEMKA. LAMCB. LCMDA. LEMDB. LAMDC. LDMEA.
LEMBE. LMECC. LAMED
- 7 LNMAA. LNMBB. LNMC. LNND. LNNE. LNMF. LNMG
- 8 LNUAA. LNUBA. LNUAB. LNUBA. LNUHB. LNUBN. LNUBB;
LNUAD. LUQAF. LFUBC
- 9 LFOAHF. LFUCOBA. LFUBOCA; LODOEFOODEF
- 10 LIRKH. LRHBA. LRHHN. LRCHA. LARCN
- 11 LRCKNH. LRCKAG. LRBCOB. LRBD
- 12 LRCKHE. LRBCOC
- 13 LJHN. LKAG. LKAOCC. LKJARB
- 14 LHJD. LHJB. LAPCC. LHFB
- 15 UAB. UBC; UCD; UEJ; UFJ; UGJ; URBCRB.
- 16 QINA; QLAC; QIAKAA; QENAAA; QRCEB; QUES; QHJAALAB;
QLCC; QUIG.
- 17 SLAA; SUAB; SLEKAA; SLAGAA; SLNMUA; SHLFOAFLPFF; SHLAALBB
- 18 LATA. LBTATA; LETKTCT; LFTCTBT; LGTG; LANTJ.
- 19 LABCTKOARJEKORJAC;
LCBATKCRJEKORJAOARJN;
LDEFUKRDRJCORJEBROFRJAOGRJN;
- 20 HJAHAVI CAY.
HJATHVILAV.
HJAVIATHAV
HJAVI KAV
HJAHAVI KCV.
HJAHAVI CVV
HJCTVIATHVVB
SLKAVKVAV
- SLAVVTYVAV.
SLAVPTTYKAVPTYV.
SLAVPTTYVATVPTTV.
MLAVTYSLVAVTVAV.
MLAVTYSLVAVTVAV
MLAVTYSLVAVTVAV
- 21 MSLMNDVYDODVCVLADCTYDUDV
TDLAVLVLMVLYLAVOV
MLAVLTVMVLYLAVOV.

EXTRATERRESTRIAL

[illegible]

31. JNVJRAVCHAV;
JNVJRBVCHAV;
JNVJRCVCHAV;
JNVJRAVJRBJRCVJRDVJREVJRFVJRCVSTVCHAV.
JNVJOAVCHAV;
JNVJOBVCHAV;
JNVJOBVCHAV;

UNCLASSIFIED

UNCLASSIFIED